Overfitting Intuition

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected model-comparison record.

Select from evidence, interpret a gap, plan a fair change and limit a diagnosis.

Keep one habit: Keep per-record errors, MAE units, split roles, versions and the untested final set visible.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Choose Informative Questions. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Types of Classification Error. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Human Classifier Simulation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Choose Informative Questions

Types of Classification Error

Human Classifier Simulation

W1 · Select using validation evidence

> Distinguish the opening record from the separate pictured example.

Your evidence record

- Opening comparison, MAE in pages: Constant training5/validation6; Line2/3; Flexible curve0/8. Use the same regression task, metric and respective split for each candidate. Lower validation error selects the development candidate; a gap alone does not establish its cause. The separate pictured Simple/Medium/Complex record uses5/6,2/3,0/7. No final-test result or training prediction rows are supplied.

- Choose among the three worked candidates and explain the **training** trap.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Interpret a training-validation gap

> Give alternative explanations and evidence to inspect.

Your evidence record

- Opening comparison, MAE in pages: Constant training5/validation6; Line2/3; Flexible curve0/8. Use the same regression task, metric and respective split for each candidate. Lower validation error selects the development candidate; a gap alone does not establish its cause. The separate pictured Simple/Medium/Complex record uses5/6,2/3,0/7. No final-test result or training prediction rows are supplied.

- Does a **training**-validation gap alone prove **overfitting**?

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Propose one controlled comparison

> Keep fitting, validation cases and the metric explicit.

Your evidence record

- Opening comparison, MAE in pages: Constant training5/validation6; Line2/3; Flexible curve0/8. Use the same regression task, metric and respective split for each candidate. Lower validation error selects the development candidate; a gap alone does not establish its cause. The separate pictured Simple/Medium/Complex record uses5/6,2/3,0/7. No final-test result or training prediction rows are supplied.

- Describe one fair next development comparison.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Limit a memorised-noise diagnosis

> Inspect provenance, coverage, grouping and label quality.

Your evidence record

- Opening comparison, MAE in pages: Constant training5/validation6; Line2/3; Flexible curve0/8. Use the same regression task, metric and respective split for each candidate. Lower validation error selects the development candidate; a gap alone does not establish its cause. The separate pictured Simple/Medium/Complex record uses5/6,2/3,0/7. No final-test result or training prediction rows are supplied.

- What extra **evidence** would you inspect before concluding a model memorized noise?

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Apply independently

> Keep one first attempt before hints or feedback.

Read and explain independently

For this A/B/C independent comparison all values are MAEs in pages. Use lowest validation MAE, then the stated simplicity tie rule; the final test is still untouched.

Models A **training** 4 validation 5, B **training** 1 validation 2, C **training** 0 validation 2. Lowest validation MAE then simpler; B is simpler than C. Choose and justify.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply independently

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Avoid title-by-title patches; improve features or thresholds that transfer.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Interpret a gap for **Overfitting** and model **capacity**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for **Overfitting** and model **capacity**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Rule A memorises eight **training** IDs and gets 8/8 **training**, 2/8 validation. Rule B uses genre and gets 6/8 **training**, 6/8 validation. Which is preferable on this **evidence**, and what test remains?

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Rule A memorises six **training** book names. It scores 6/6 familiar and 2/6 new. Rule B uses relevant traits but is untested.

What does the **evidence** support?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ Extra practice · A changed case

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

Rule A memorises six **training** book IDs. Rule B uses a relevant topic feature. Why might A fail on a new ID? What test is needed?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 16 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Rule A memorises eight **training** IDs and gets 8/8 **training**, 2/8 validation. Rule B uses genre and gets 6/8 **training**, 6/8 validation. Which is preferable on this **evidence**, and what test remains?

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Models A **training** 4 validation 5, B **training** 1 validation 2, C **training** 0 validation 2. Lowest validation MAE then simpler; B is simpler than C. Choose and justify.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

> Keep the explanation and the support you used.

Use this when helpful

A fictional project compares three book-length **prediction** models on the same development sets.

Mean absolute errors in pages: Simple **training** 5, validation 6; Medium **training** 2, validation 3; Complex **training** 0, validation 7.

Select lowest validation error; untouched final tests are not used for selection.

AI ACADEMY / AI-2/16

Training and validation errors

	Training MAE pages	Validation MAE pages
Simple	5	6
Medium	2	3
Complex	0	7

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/16

Overfitting Intuition

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Training comparison

Complex fits training perfectly with zero error.

↓

2. Validation comparison

Medium has error 3, versus Simple 6 and Complex 7.

↓

3. Choose for development

Select Medium under the declared lowest-validation-error rule.

↓

4. Final evaluation

Freeze the selection, then evaluate new reserved cases.

CHANGE ONE THING

Complex's validation error becomes 2 on a newly documented, fair development comparison.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Does zero **training** error justify choosing Complex in the original record?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Compare recommender revisions on fixed development cases, reserving new cases for the final presentation.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Transparent Book Recommender

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Compare recommender revisions on fixed development cases, reserving new cases for the final presentation.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional project compares three book-length **prediction** models on the same development sets.
- Which labelled step matches this detail? Freeze the selection, then evaluate new reserved cases.
- 1. Final evaluation
- 2. **Training** comparison
- 3. Choose for development
- 4. Validation comparison

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Final evaluation Freeze the selection, then evaluate new reserved cases. A flexible model can fit quirks of its **training** set without predicting new examples well.

Here Complex's **training** success occurs alongside the worst validation error. The numbers are invented comparisons, not a claim that **complexity** always hurts.

Choosing Medium uses development **evidence**; its validation score must not be advertised as an untouched final-test result.

Changed-case question

- Changed-case practice: Complex's validation error becomes 2 on a newly documented, fair development comparison. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. The stated rule now selects Complex.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: The stated rule now selects Complex. The rule compares validation errors; it does not ban complex models.

Repair and save your work

Review Choose Informative Questions (ai-2/15). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Choose the best original question and explain why its more balanced split matters here..

Return to this week and explain what you changed.

Review Types of Classification Error (ai-2/13). Goal: Build a labelled confusion matrix from matched reference and **prediction** pairs. Short practice: Which error causes a sound book to be sent for unnecessary repair review?.

Return to this week and explain what you changed.

Review Human Classifier Simulation (ai-2/10). Goal: Construct a classifier with a stated **input** format, selected rule and recorded **output**.

Short practice: Why are J and M initially treated differently even though both are illustrated mysteries?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

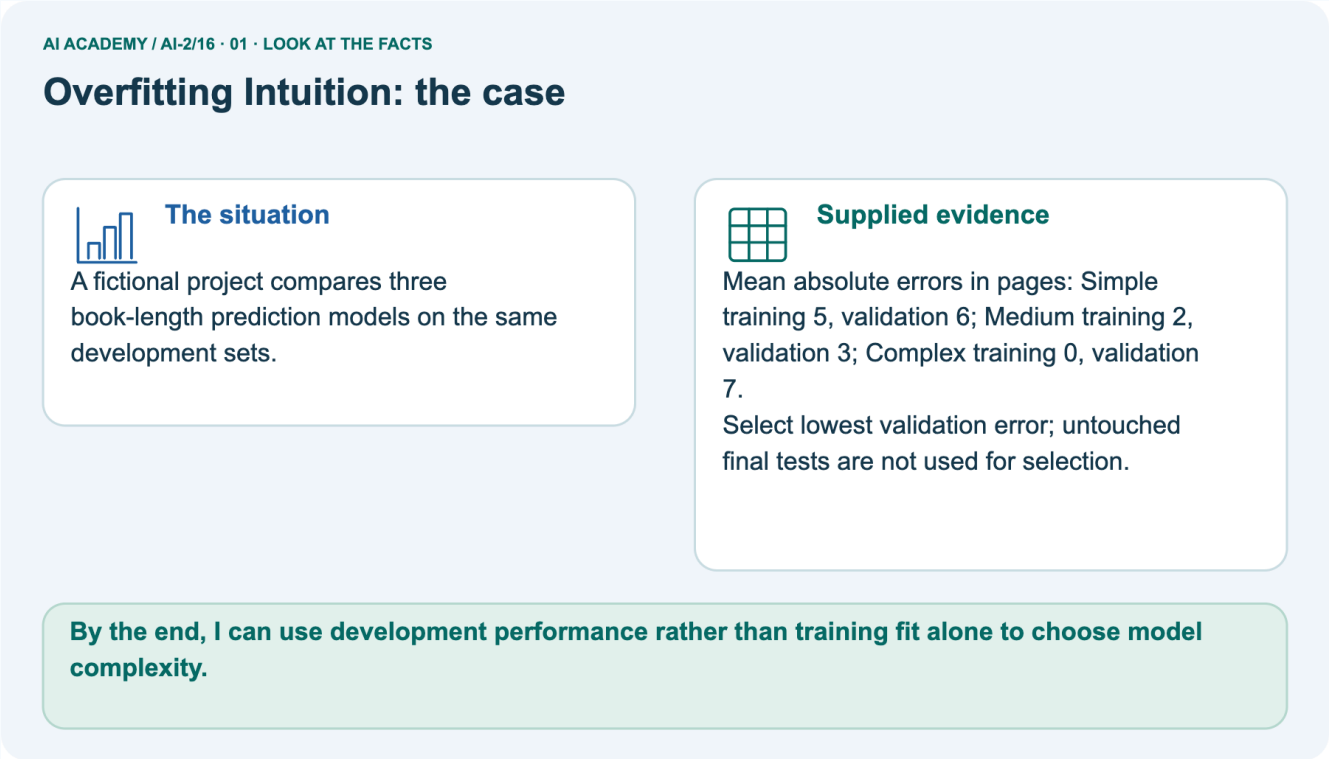
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Overfitting Intuition: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional project compares three book-length **prediction** models on the same development sets.

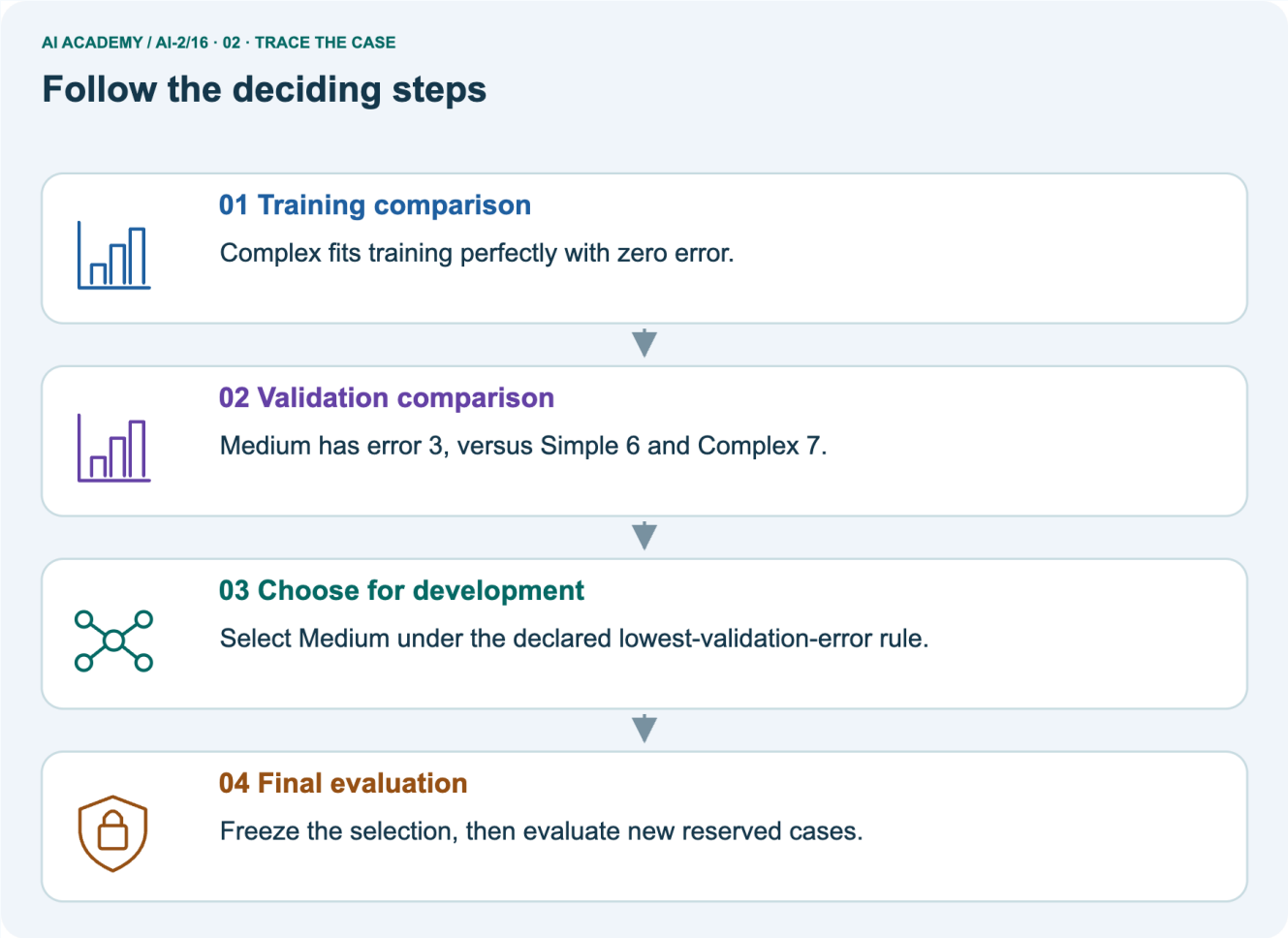
Supplied facts: Mean absolute errors in pages: Simple **training** 5, validation 6; Medium **training** 2, validation 3; Complex **training** 0, validation 7.

Select lowest validation error; untouched final tests are not used for selection.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



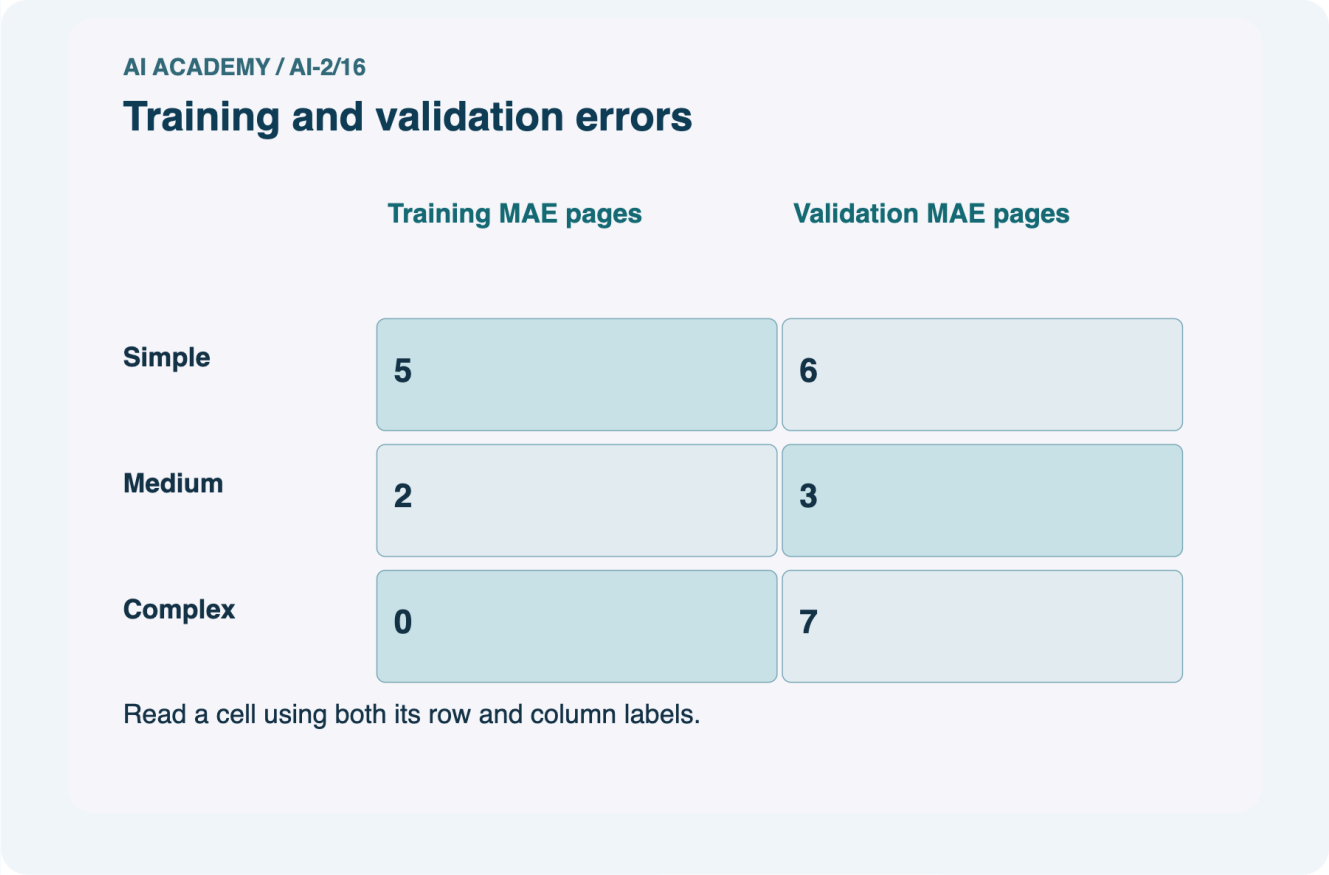
Follow the labelled picture with the matching explanation. Open the vector diagram

Training comparison: Complex fits **training** perfectly with zero error. → Validation comparison: Medium has error 3, versus Simple 6 and Complex 7. → Choose for development: Select Medium under the declared lowest-validation-error rule. → Final evaluation: Freeze the selection, then evaluate new reserved cases.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Training and validation errors



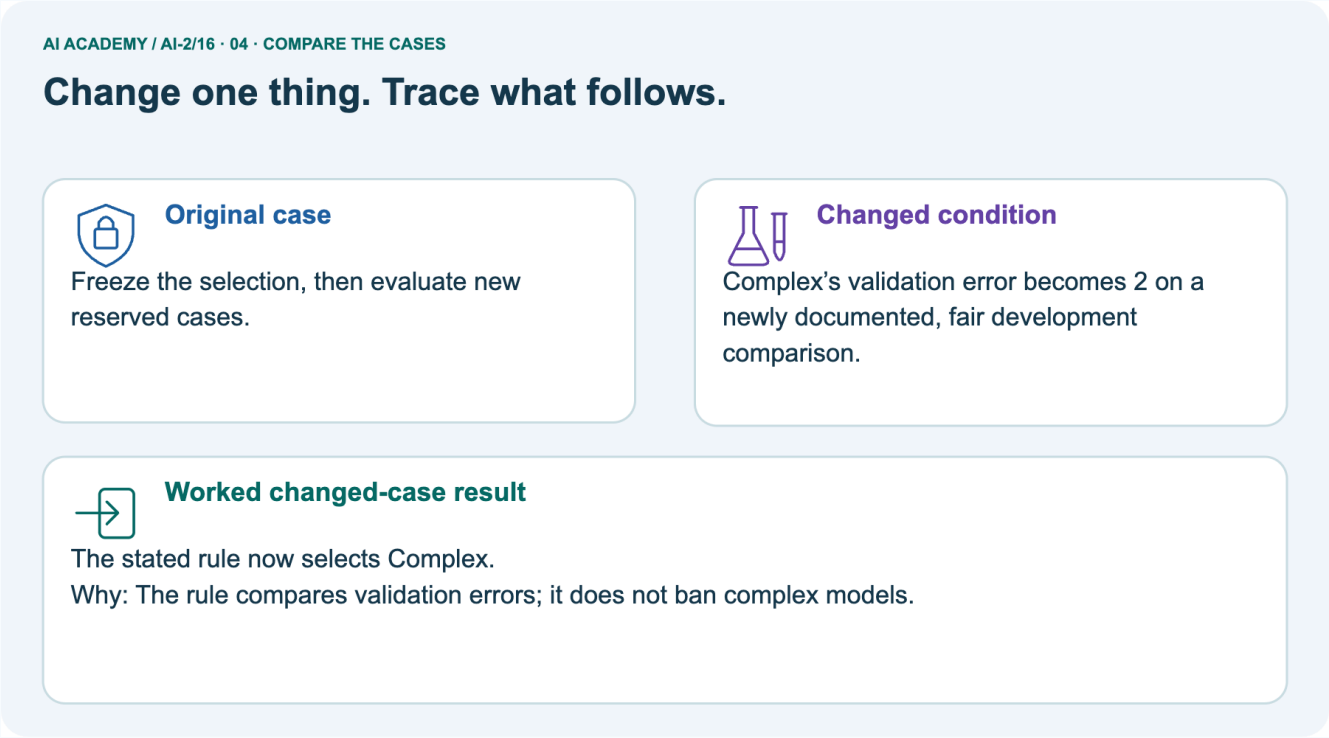
Follow the labelled picture with the matching explanation. Open the vector diagram

Training and validation errors. Simple: **Training** MAE pages = 5, Validation MAE pages = 6. Medium: **Training** MAE pages = 2, Validation MAE pages = 3. Complex: **Training** MAE pages = 0, Validation MAE pages = 7.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

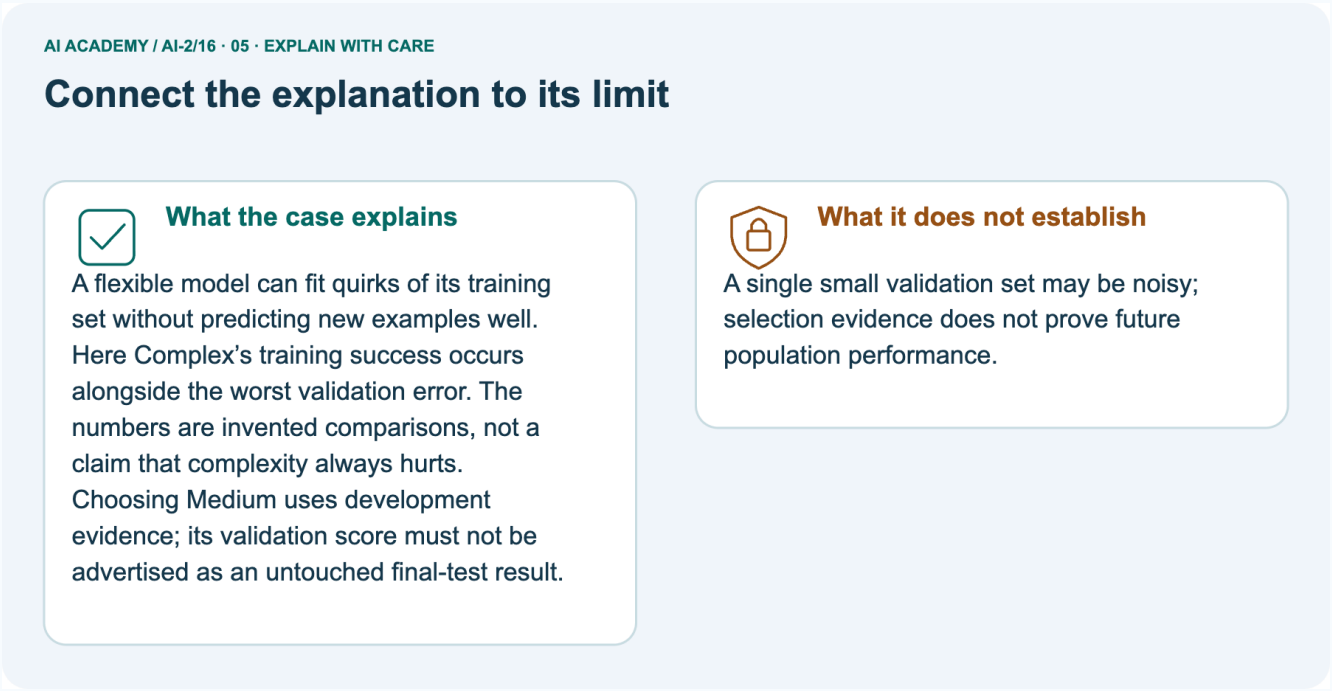
Original case: Freeze the selection, then evaluate new reserved cases. Changed condition: Complex's validation error becomes 2 on a newly documented, fair development comparison.

Expected result: The stated rule now selects Complex. Reason: The rule compares validation errors; it does not ban complex models.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A flexible model can fit quirks of its **training** set without predicting new examples well. Here Complex's **training** success occurs alongside the worst validation error.

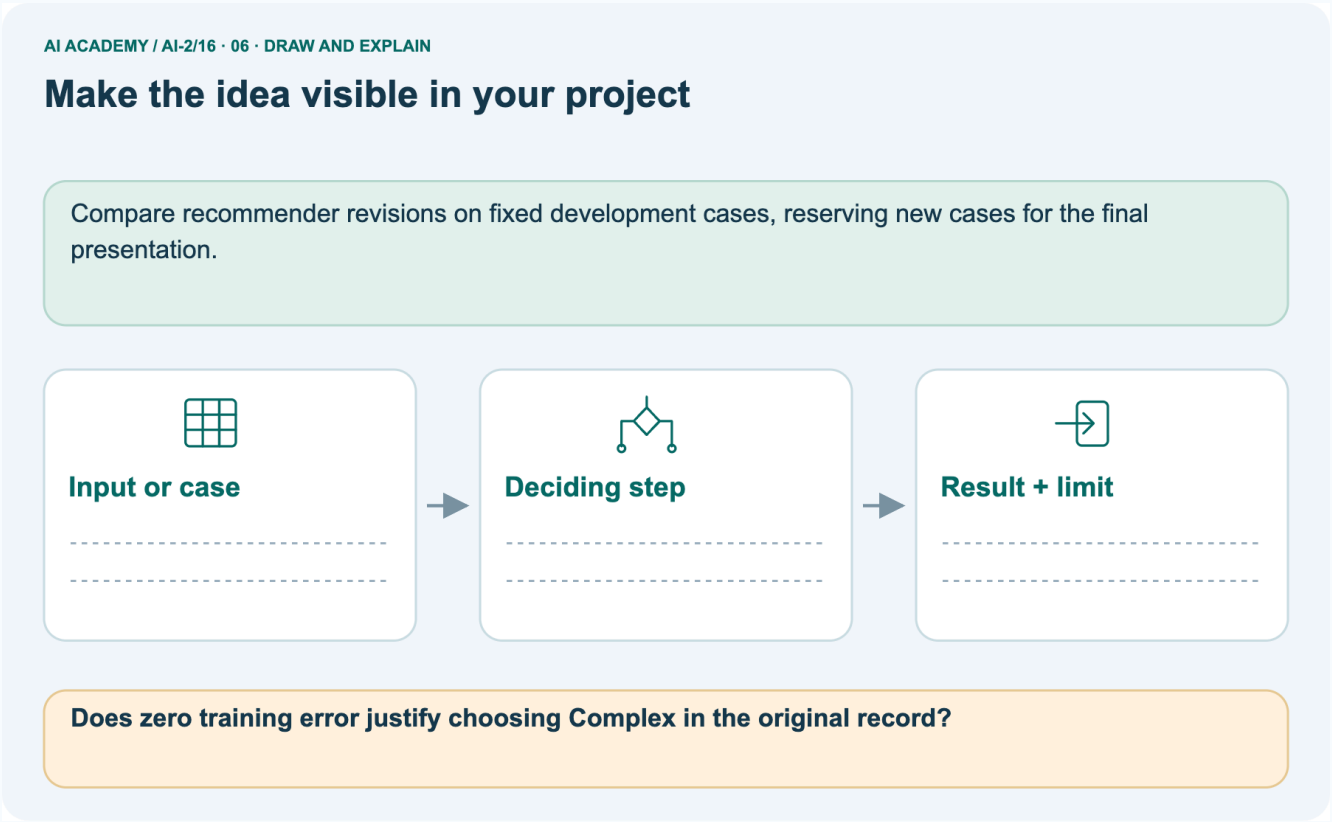
The numbers are invented comparisons, not a claim that **complexity** always hurts. Choosing Medium uses development **evidence**; its validation score must not be advertised as an untouched final-test result.

Limit: A single small validation set may be noisy; selection **evidence** does not prove future population performance.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Compare recommender revisions on fixed development cases, reserving new cases for the final presentation. Draw your **input**, deciding step, result and limit.

Supported question: Does zero **training** error justify choosing Complex in the original record?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Overfitting Intuition: the case

02. Follow the deciding steps

03. Training and validation errors

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.